Toh Wei Xuan

Student in Computer Science Real-Time Interactive Simulation

2401687@sit.singaporetech.edu.sg · +65 82995285 · Singapore

EDUCATION

Real-Time Interactive Simulation, Computer Science

September 2024

Singapore Institute of Technology

PROFILE

Computer Science student specializing in Real-Time Interactive Simulation with experience in C++ game development, gameplay systems, AI behavior, and real-time simulation. Former Air Force Engineer with four years of experience in a high-reliability engineering environment, including troubleshooting, quality control, process improvement, and team leadership. Strong problem-solving and technical skills with experience working collaboratively on software projects.

EXPERIENCE

Air Force Engineer

August 2019 - August 2023

Republic of Singapore Air Force (RSAF)

Singapore

- Maintained the serviceability of fighter aircraft within a high-reliability engineering environment.
- Applied systematic troubleshooting and problem-solving to aircraft maintenance activities.
- Implemented new procedures and technologies that improved efficiency and streamlined operations.
- Led a team to achieve expected maintenance goals while reducing human-factor errors.
- Applied quality control, technical procedures, and effective team communication to maintain operational standards.

TECHNICAL SKILLS

Programming:
C++

Software Engineering:
Object-Oriented Programming
ECS Architecture
Debugging
Event-Driven Systems

Professional:
Team Collaboration
Problem Solving
Technical Documentation
Technical reporting

Game Development:
Real-Time Simulation
Gameplay Programming
Game Systems

Technical:
Collision Detection
AI/Steering Behavior
Scene Management
UI Integration

PROJECTS

Soot & Scoot — Game Development Project

GAM 200/250 | C++ | Custom Game Engine | Team Project | Software Engineer |

- Developed gameplay systems in C++ within a custom game engine using an Entity Component System (ECS) architecture.
- Implemented collision detection, event handling, scene management, and game-state systems.
- Developed steering and AI behaviour, including target-seeking and obstacle avoidance for in-game entities.
- Integrated gameplay systems with menus, UI interactions, and scene transitions.
- Debugged, tested, and refined systems through iterative development and team collaboration.

LANGUAGES

English: Fluent

Chinese: Native